

THE STRUCTURE OF MATHEMATICAL REALITY

Phase IV — Chapter 12

Information Theory and the Cost of Description

Note Template

Self-Study Curriculum

How to use this template

Complete these notes **after** the reading and the lecture. The goal is not to copy phrases from the text — it is to reconstruct the ideas in your own language. Leave every item blank on first pass; fill it in from memory, then check.

1. Information = _____ (in Shannon's framework)

The information content of an event with probability p is:

2. Entropy measures (three equivalent interpretations)

- a.
- b.
- c.

3. Shannon's source coding theorem

your words, what is the hard floor?

4. KL divergence measures

In modeling terms, it is the cost of:

5. Mutual information measures

How it differs from correlation:

6. The connection between compression and modeling

trace the chain: model $\rightarrow$ compression $\rightarrow$ entropy $\rightarrow$ rules

7. The difference between statistical compression and generative

compression:

Statistical:

Generative:

Why generative is more powerful:

8. Where I see these ideas in my own work

entropy, KL divergence, cross-entropy loss, where do they appear and what are they telling me?
